

RAGHUL SHREERAM SIVAKUMARAN

✉ rsivakumaran@ucsd.edu ☎ +1 858-319-7035 🔗 [linkedin.com/in/Raghul](https://www.linkedin.com/in/Raghul) 🐙 github.com/RaghulShreeramS

Education

University of California San Diego (UCSD)

Sept'23 – Present

Master of Science in Mechanical Engineering (Robotics)

Indian Institute of Technology, Madras (IIT Madras)

Aug'18 – May'23

BTech in Engineering Design and MTech in Automotive Engineering

Relevant Coursework

Introduction to Robotics — Sensing and Estimation in Robotics — Planning and Learning in Robotics — Reinforcement Learning — Machine Learning for Physical Applications — Introduction to Field and Service Robotics — Pattern Recognition and Machine Learning — Fundamentals of Automotive Systems — Control of Automotive Systems — Vehicle Dynamics

Technical Skills

Languages and Tools: Python, C/C++, MATLAB, Pytorch, ONNX, TensorRT, Numpy, Pandas, Git

Simulation: ROS, MuJoCo, Pybullet, Coppelia Sim, Simulink, Simscape, Fusion360

Experience

SIEMENS – Autonomous Software Intern R&D

Dec'21 – May'22

Autopilot Team

Bengaluru, India

- Simulated multi-vehicle collisions using **Pybullet**, varying collision angles and velocities, to extract dissipation energy.
- Collected over **350 collision** instances and developed a **Neural Network** with **4 hidden layers** using ReLU activation to predict dissipation energy and map it to crash severity. Achieved **96.52%** accuracy after optimizing hyperparameters

Graduate Research Assistant - Existential Robotics Lab, UCSD

Jun'24 – Present

Guide: Dr. Nikolay Atanasov, Director of Existential Robotics Laboratory, UCSD

San Diego, CA

- Developed a **reinforcement learning** framework in **MuJoCo**, where an F1tenth car navigates through randomly placed obstacles. The simulation environment incorporated both **visual** inputs from a Intel Real Sense stereo camera and **state** observables.
- Designed and implemented a **MultiInputPolicy** using **CNN** for visual inputs and **MLP** for state inputs. Trained the model using Proximal Policy Optimization (**PPO**) algorithm to predict **steering angles** and **throttle** for the car.
- Achieved successful navigation after **500K** timesteps by **parallelizing** learning across **8 environments**, designing a tailored reward function and fine-tuning hyperparameters for optimal performance.
- Trained and deployed a **PyTorch** model by converting it to **TensorRT** with **ONNX**, building **ROS** nodes, and executing on the **Jetson TX2 GPU** using **CUDA**, enabling successful navigation in the F1tenth car.

Undergraduate Research – IIT Madras, India

Aug'22 – Dec'22

Guide: Prof. C S Shankar Ram, Engineering Design, IIT Madras

Chennai, India

- **Machine Learning for Vehicle Predictive Maintenance:** Analyzed temperature's impact on air brake systems, pushrod stroke, using wheel speed and temperature data. Implemented advanced Neural Networks for pushrod stroke prediction and developed a fault diagnostic scheme, enhancing system fault detection and isolation accuracy.

Projects

Sense-Plan-Act on a Real Robot – Introduction to Robotics

Sep'23 – Dec'23

Course Instructor: Dr. Henrik I. Christensen, Director, Contextual Robotics Institute, UCSD

San Diego, CA

- **Calibrated** mecanum wheels and implemented a **PID** controller for accurate waypoint navigation and straight-line movement on a **Qualcomm RB5 platform** using Robot Operating Software (**ROS**).
- Applied **SLAM** using **depth data** from a camera, with **A*** algorithm for efficient navigation and obstacle avoidance.
- Developed a **Roomba-like** robot that mapped and navigated its environment autonomously using SLAM for coverage.

Autonomous Navigation with obstacle avoidance – ML for Physical Applications

Apr'24 – Jun'24

Course Instructor: Dr. Yuanyuan Shi, ECE, UCSD

San Diego, CA

- Used **Imitation learning** to train a **CNN** model that maps **visual data** from a Husky robot's onboard camera to **keyboard inputs**, enabling autonomous navigation and obstacle avoidance in a simulated **PyBullet** environment.

Visual - Inertial SLAM – Sensing and Estimation in Robotics

Jan'24 – Mar'24

Course Instructor: Dr. Nikolay Atanasov, Director of Existential Robotics Laboratory, UCSD

San Diego, CA

- Implemented an Extended Kalman Filter (**EKF**) to predict and update the state at each timestep upon detecting **landmarks**, while utilizing kinematic equations and **IMU** measurements for pose estimation over time.

Planning and Learning in Robotics

Apr'24 – Jun'24

Course Instructor: Dr. Nikolay Atanasov, Director of Existential Robotics Laboratory, UCSD

San Diego, CA

- Implemented a **Dynamic Programming** based algorithm to solve the Door-Key problem, a pathfinding challenge within a grid-based simulation environment.

Mobile Manipulation - Robotics

Jan'24 - Mar'24

Course Instructor: Dr. Michael T. Tolley, MAE, UCSD

San Diego, CA

- Developed a mobile manipulator simulation in MATLAB, incorporating kinematics, **PID** control algorithms, and trajectory generation, integrated with **CoppeliaSim** for cube picking and dropping tasks.